4MTH 311

Lecture 8

Isomorphism Theorems

Theorem: For any subgroup H of a group G, the $H^2 = H$ where H^2 denotes the product HH.

Definition: A group G is simple if

$$N \trianglelefteq G \Rightarrow N = \{e\} \quad \text{or} \quad N = G.$$

Quotient Groups

Theorem: Let $N \trianglelefteq G$. Then the operation

$$(xN)(yN) := (xy)N$$

is well-defined and gives a group structure of

G/N. ← quotient group.

Proof

$\boxed{1}$ Let $x_1 N = x_2 N$ and $y_1 N = y_2 N$. Then

$x_1^{-1} x_2 \in N$, $y_1^{-1} y_2 \in N$.

We want to show that

$$(x_1 y_1)N = (x_2 y_2)N$$

Compute

$$(x_1 y_1)^{-1}(x_2 y_2) = y_1^{-1} x_1^{-1} x_2 y_2 \in N$$

Hence the operation is well defined.

$\boxed{2}$ Let xN and yN is in G/N.

(i) $(xN)(yN) = x(Ny)N$

$= x(yN)N$ Since N is normal

$= (xy)\cdot N \cdot N$

$= xy\,N$ Since $N^2 = N$.

There G/H is closed

(ii) The identity element is

$$eN = N$$

Check (eg. $(eN)(xN)$)

(iii) For $x \in G/N$, define

$$(xN)^{-1} = x^{-1}N$$

Check $((xN)(x^{-1}N))$

Therefore G/N is a group.

Remark: You have a group G and $N \trianglelefteq G$.

- The map

$\phi : G \longrightarrow G/N$ is called the quotient homomorphism defined by

$$x \longmapsto xN$$

Key Fact

$$\ker(\phi) = N$$

The kernel is the set of all elements that maps to the identity in G/N. The identity in G/N is N itself. So

$$\phi(x) = N \iff x \in N.$$

The First Isomorphism Theorem

Let $\phi : G \to H$ be a homomorphism. Then the map

$\psi : G/\ker(\phi) \longrightarrow \operatorname{Im}(\phi)$ defined by

$$x\ker(\phi) \longmapsto \phi(x)$$

is a well defined isomorphism.

- Tells you what homomorphisms does to a group
- Understands homomorphisms via quotients

Third Isomorphism Theorem

Let $N \trianglelefteq G$. Then there is a natural bijection

$$\left\{ \begin{array}{c} \text{Subgroups of } G \\ \text{containing } N \end{array} \right\} \longrightarrow \left\{ \text{Subgroups of } G/N \right\}$$

$$H \longrightarrow H/N = \{ xN \mid x \in H \}.$$

Moreover $H/N \trianglelefteq G/N \iff H \trianglelefteq G$ and in this case

$$G/H \cong \frac{G/N}{H/N}.$$